

Dr Thomas Bartsch
Department of Mathematical Sciences
Loughborough University
Loughborough, LE 11 3TU
UK



To whom it may concern

Loughborough, 29 July 2025

Letter of reference for Christopher Bruce

Dear Madam, dear Sir,

I am writing to support the application of Christopher Bruce for doctoral studies.

Christy studied here in Loughborough for a BSc in Natural Sciences from October 2019 to summer 2022, when he completed with first class honours, and subsequently for an MSc in Financial Mathematics, which he achieved with distinction. I taught him in a third-year undergraduate module on Graph Theory and later supervised his MSc thesis.

During his MSc studies, Christy approached me with a proposal for a project that he had designed himself: He was planning to investigate the impact of the Covid pandemic on the US stock market with tools taken from the theory of complex networks. By the time we discussed the proposal, Christy had collected the relevant data and taken first steps of analysis in Python. Over the following months he completed an outstanding thesis for which he received the exceptional mark of 93%. He approaches his problem from several different points of view and uses a range of graph theoretical concepts. He has since further elaborated his analysis. We are currently preparing the material for publication.

Christy brought a lot of creativity and enthusiasm to his work. He had lots of ideas for interesting research questions to ask of his data and possible methods of analysis. I supported him in shaping his ideas into a realistic project and in the choice of suitable mathematical tools. Nevertheless, his project is, in essence, entirely his own.

It was a pleasure to work with Christy; I would happily do that again. I have no doubt that he has the ability to succeed in doctoral studies and to complete an excellent project. I warmly recommend him for admission.

Yours sincerely,

Dr. Thomas Bartsch

Dr Thomas Bartsch
Senior Lecturer in Applied Mathematics